

# Customer Shopping Behavior Analysis

This project analyzes 3,900 purchases across multiple product categories to uncover spending patterns, customer segments, product preferences, subscription behavior, and actionable recommendations to guide strategy.



# Project Scope & Tools



## Data & Analysis

3,900 rows × 18 columns.  
Focus: customer demographics, purchase details, and shopping behavior.



## Database & SQL

PostgreSQL for structured analysis; SQL used for business transactions and segmentation queries.



## Visualization & Reporting

Power BI for interactive dashboards;  
PowerPoint/Excel for reporting and storytelling.

# Dataset Summary

Rows: 3,900. Columns: 18. Key features include customer demographics (age, gender, location, subscription status), purchase details (item, category, amount, season, size, color), and behavior (discounts, promo codes, previous purchases, frequency, review rating, shipping type).

**i** 37 missing values found in Review Rating; treatment: median imputation by product category.

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|------|------|----------------|---------|-------------|-------|---------|-------|-------|-------|--------|--|
| Chae | To   | Dp Caotionna   | lan     | Yiai        | IG    | EGAO    | R31   | 1101  | L61   |        |  |
| 3.9  | 2.33 | S4C55125       | 2.5.998 |             | 15.50 | 8.90    | 9.93  | 9.11  |       |        |  |
| 6.0  | 2.52 | S4C88303       | 3.90    |             | 8.29  | 8.00    | 3.90  |       |       |        |  |
| 3.9  | 1.33 | S6C25245       | 3.33    |             | 81.99 | 8.00    | 1.84  | 3.34  |       |        |  |
| 210  | 2.42 | K6C98300       | 2.3.005 |             | 5     | 81.48   | 6.50  | 4.40  | 3.41  |        |  |
| 833  | 2.35 | 46C16683       | 3.1.000 |             | 2     | 81.99   | 1.90  | 2.30  | 3.10  |        |  |
| 905  | 2.78 | S6C20000       | 4.2.200 |             | 2     | 21.85   | 6.50  | 3.30  | 6.10  |        |  |
| 607  | 1.20 | SGC28088       | 2.6.200 |             |       | 51.90   | 6.50  | 1.00  | 3.10  |        |  |
| 308  | 1.38 | A6C78277       | 82.500  |             | 8     | 4.36    | 1.30  | 1.90  |       |        |  |
| 200  | 2.59 | S6C75183       | 83.590  |             | 51    | 61.57   | 1.50  | 1.18  |       |        |  |
| 809  | 1.89 | S6C53199       | 38.900  |             |       | 91.75   | 1.40  | 1.18  | 1.18  |        |  |
| 100  | 1.24 | Kac23344(pri)  | 181.200 |             | 87    | 71.76   | 0.00  | 7.50  | 1510  |        |  |
| 808  | 1.38 | S0C33005       | 182.200 |             |       | 83.88   | 8.90  | 6.80  |       |        |  |
| 503  | 1.42 | SI40011        | 31.50   |             | 1     | 81.75   | 5.00  | 8.85  | 52.9  |        |  |
| 190  | 1.31 | SeC92386       |         |             |       | 81.13   | 5.00  | 1.18  | 41.8  |        |  |
| 198  | 1.33 | 6.6639001      | 811.908 |             | 8     | 55.58   | 5.90  | 8.80  | 6.10  |        |  |
| 190  | 1.31 | G6430008       | 539.500 |             | 7     | 31.53   | 2.80  | 6.90  |       |        |  |
| 209  | 1.32 | 6eC27138       | 888.305 |             | 81    | 81.38   | 6.00  | 6.35  | 3.10  |        |  |
| 828  | 1.94 | N6C35589       | 888.500 |             |       | 55.99   | 7.00  | 6.06  | 6.08  |        |  |
| 132  |      | Ans31108       |         |             |       |         |       |       |       |        |  |
| 398  |      | Dec22102       | 15.9    |             | 30    | 58.50   | 5.90  | 6.56  | 37.4  |        |  |
| 596  | 3.3  | Dec22102       | 9.500   |             | 30    | 12.70   | 1.50  | 1.30  | 8.30  |        |  |
| 309  | 1081 | Eac33328       |         |             |       | 35.71   | 8.90  | 8.59  |       |        |  |
| 059  | 585  | Eec25488       | 5.38    |             |       | 41.13   | 0.85  | 8.87  | 2.87  |        |  |
| 095  | 153  | Sac37187       | 8.509   |             | 1     | 25.13   | 6.80  | 4.59  | 1.14  |        |  |
| 033  | 174  | Dac26028       | 5.50    |             |       | 13.56   | 8.00  | 1.23  | 1.18  |        |  |
| 209  | 228  | Sec30934       | 5.58    |             | 5     | 11.14   | 6.83  | 1.93  |       |        |  |
| 108  | 174  | Gac70238       | 6.90    |             | 1     | 11.15   | 11.83 | 9.50  | 1.42  |        |  |
| 108  | 127  | Sec70956       | 67.00   |             |       |         |       |       |       |        |  |
| 208  | 508  | Gac70228       | 2.80    |             | 5     | 81.80   | 11.83 | 1.73  | 1.53  |        |  |
| 158  | 152  | Ger28245       | 85.55   |             | 3     | 51.30   | 8.81  | 2.88  | 1.95  |        |  |
| 909  | 123  | Sec15695       |         |             | 18    | 51.74   | 6.50  | 1.70  | 2.50  |        |  |
| 000  | 112  | Eac118464      | 17.95   |             | 21    | 51.00   | 11.00 | 1.15  | 2.17  |        |  |
| 108  | 133  | Sec26228       | 3.51    |             | 3     | 81.39   | 11.83 | 2.20  | 1.44  |        |  |
| 304  | 119  | Sec110210      | 2.37    |             | 3     | 91.01   | 11.00 | 5.29  |       |        |  |
| 070  | 158  | Gac78C38       | 2.59    |             | 1     | 31.03   | 1.23  | 3.70  | 2.38  |        |  |
| 190  | 217  | Eac212138      | 58.80   |             | 1     | 835     | 1084  | 8.35  |       |        |  |
| 200  | 834  | Eac239182      | 20.90   |             | 2     | 859     | 1.90  | 1.99  | 5.58  |        |  |
| 598  | 325  | Sec309145      | 12.15   |             | 5     | 899     | 1.01  | 2.74  |       |        |  |
| 558  | 193  | dah29184       | 10.80   |             | 1     | 840     | 1.35  | 1.08  | 2.00  |        |  |
| 703  | 292  | Dac10008       | 0.0     |             | 0     | 539     | 1.03  | 8.00  | 3.50  |        |  |
| 883  | 313  | Dac190025      | 8.9     |             | 5     | 887     | 2.90  | 3.04  |       |        |  |
| 93   | 113  | Eac25040       | 10.60   |             | 4     | 888     | 2.05  | 1.08  | 11.58 |        |  |
| 99   | 423  | Nac23878       | 18.90   |             | 3     | 880     | 3.80  | 2.26  | 8.35  |        |  |
| 113  |      | Gac510.20(pri) | 9.9     |             | 3     | 546     | 3.00  | 1.08  | 11.49 |        |  |
| 113  |      | Sec313133      | 58.90   |             | 3     | 889     | 5.00  | 8.23  | 3.29  |        |  |
| 201  |      | Lac282.38      |         |             | 8     | 838     | 6.00  | 8.09  | 8.39  |        |  |
| 124  |      | Sec2152.53     | 69.40   |             | 5     | 899     | 5.00  | 1.51  |       |        |  |
| 96   |      | Sec2309(pri)   | 39.80   |             | 2     | 310     | 9.00  | 3.83  | 21.55 |        |  |
|      |      | Dite22133      | 22.88   |             | 5     | 335     | 9.00  | 8.30  | 8.33  |        |  |
|      |      | Rac18648       |         |             |       | 338     | 3.30  | 1.13  | 8.54  |        |  |
|      |      | Rac23818       | 38.90   |             | 51    | 310     | 3.95  | 1.45  | 15.49 |        |  |



# Exploratory Data Analysis — Python Workflow

01

## Data Preparation

Loaded with Pandas; `df.info()`, `df.describe()`, `value_counts` used to profile the data.

02

## Cleaning & Imputation

37 missing review ratings imputed using category median; duplicates and inconsistencies checked.

03

## Feature Engineering

Created `age_group` and other derived variables to support segmentation and behavioral analysis.

04

## Database Integration

Standardized column names to `snake_case` and loaded cleaned data to PostgreSQL via SQLAlchemy.



# Key SQL Findings — Revenue & Behavior



## Revenue by Gender

Female: 75,191 USD; Male: 157,890 USD — clear revenue split by gender in the dataset.



## High-Spending Discount Users

839 rows identified where customers used discounts yet spent above the average purchase amount.



## Top Rated Products

Top 5 by average rating: Gloves 3.86; Sandals 3.84; Boots 3.82; Hat 3.80; Skirt 3.78.





# Additional SQL Insights

## Shipping & Subscriptions

Average purchase: Standard = 58.46 USD; Express = 60.48 USD.  
Subscribers: 1,053 customers (avg spend 59.49, total revenue 62,645).  
Non-subscribers: 2,847 customers (avg spend 59.87, total revenue 170,436).

## Customer Segments & Revenue

Segments: Loyal 3,116; New 83; Returning 701. Revenue by age\_group: Young Adult 62,143; Middle-aged 59,197; Adult 55,978; Senior 55,763 — spending varies by age cohort.

# Product & Category Highlights



## Top Category

Clothing is the highest revenue-generating category and also records the highest sales volume.



## Top Items by Volume

Category leaders: Accessories — Jewelry (171 orders), Sunglasses (161), Belt (161).  
Clothing — Blouse (171), Pants (171), Shirt (169).



## Discount-Dependent Products

Highest discount rates: Hat 50.00%; Sneakers 49.66%; Coat 49.07%; Sweater 48.17%; Pants 47.37% — discounts drive purchases for these items.

# Dashboard & Visualizations (Power BI)

An interactive Power BI dashboard was developed to visualize customer behavior and business performance. Components include KPI cards (Total Customers, Average Purchase Amount, Average Review Rating), interactive filters (Gender, Category, Subscription Status, Season, Shipping Type), and visualizations like Revenue by Category, Sales by Category, Revenue by Age Group, and Subscription Distribution.





# Key Insights

## Revenue Drivers

Clothing leads revenue and volume; certain high-rated products show strong preference and steady purchases.

## Subscription Opportunity

Majority are non-subscribers — opportunity to increase adoption through targeted offers and faster shipping perks.

## Repeat Customers

Repeat buyers show stronger purchasing behavior and contribute disproportionately to performance.

## Discount Influence

Discount campaigns significantly affect purchase decisions for specific products and segments.



# Business Recommendations & Conclusion

Recommendations: increase subscription adoption via exclusive discounts, loyalty rewards, and faster shipping; strengthen tiered loyalty programs for repeat buyers; optimize discount strategies focused on high-conversion products and segments; promote top-selling, highly rated products in marketing; target high-value age groups with tailored campaigns; and use personalized recommendations to convert first-time buyers into repeat customers.

Conclusion: The project demonstrates an end-to-end analytics workflow—data cleaning, feature engineering, SQL analysis, dashboarding, and reporting—turning 3,900 transactions into actionable insights that inform revenue growth, retention, and product promotion strategies.



## Next Steps

Implement targeted subscription campaigns, A/B test discount mechanics, and expand the Power BI dashboard to track impact over time.



## Deliverables

Cleaned dataset in PostgreSQL, reproducible Python notebooks, SQL queries, and interactive Power BI dashboard for stakeholders.